NILAKSHI ROY
Data Scientist
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Portfolio: roynilakshi91.github.io/Nilakshi_Roy_portfolio

WORK EXPERIENCE

TikTok USDS, Data Science and Analytics, San Jose, CA Aug 2022 – Present

- Built predictive ML models (CatBoost) on user behavior and account-level signals to mitigate abnormal fund flow risk, **reducing user complaints from 1,400 to 690/day and cutting chargeback rate from 4% to 1.6% within 5 weeks**.
- Systematized pre-launch performance evaluations using A/B testing experimental design and champion-challenger frameworks to optimize decision thresholds and minimize **false-positive rates to less than 0.01%.**
- Developed Trust Layer ML solutions using TF-IDF text vectorization with gradient boosting to generate confidence scores, enabling trusted requests to be fast-tracked and ensuring VIP user experience, driving **$11K/day in incremental revenue.**
- Supported U.S. campaign initiatives using cohort and funnel drop-off analyses, **A/B testing**, and **causal inference** to measure impact, and applied an ML model to identify coordinated activity, trend manipulation, and inauthentic engagement, achieving a **94% campaign success rate (**20% increase in user participation, avg chargeback rate below 1.3%)**.**
- **Prevented $1.7M in chargeback losses** by leveraging LLM-based content classification model to identify abnormal live rooms and building rule-based detection model to flag suspicious recharge, gifting behavior, and coordinated cross-regional bot activity.
- Built agentic AI workflows using Claude and MCP to automate data retrieval, weekly performance analysis, metric monitoring and regression checks , and reporting, **reducing manual effort by 86% and saving 15+ hours per week.**
- Architected an automated fake-creator detection and enforcement pipeline using registration, content, and behavioral signals, integrating LLM-based classification model to **blacklist 650 creators** and **block 12K fake registrations per week**, while maintaining nearly **100% precision.**
- Established the **end-to-end analytics and ML framework** for TikTok's PGC (Professionally Generated Content) business, covering data pipelines, metric definition, model development, experimentation, and cross-functional stakeholder coordination.
- Conducted exploratory and statistical analysis across large-scale datasets and built **20+ interactive dashboards** with real-time alerts for metric monitoring and data quality, while collaborating with cross-functional stakeholders.

Google, Data Specialist, Mountain View, CA (*Contract)* May 2021 – Aug 2022

- Optimized reporting dashboards and planning tools, **reducing data processing time by 80%** and implementing automation through **ETL scripts** to cut weekly manual labor.
- Managed data integrity and account mapping within cloud infrastructure platforms like Salesforce; developed **SQL** test scripts to automate EDA and KPI analysis.
- Translated complex datasets into actionable insights for the GCP Sales operations team, supporting GTM strategy and success through advanced data modeling and creating weekly business reports (WBRs) to ensure systematized data delivery and performance tracking.

Apple Inc., Data Reporting and Testing Analyst, Sunnyvale, CA (Contract) May 2020 – Mar 2021

- Analyzed pre- and post-release reports for iOS systems to identify performance inefficiencies and support risk mitigation efforts for global product launches.
- Leveraged **SQL** to extract and analyze server-side data, developed data-modeling flowcharts for key dimensions and product performance metrics, and built **KPI dashboards** and daily performance reports to communicate insights and support cross-functional stand-ups.

American International Group (AIG), Data Analyst, Bengaluru, India Sep 2016 – Apr 2019

- Analyzed large-scale datasets within financial services to identify patterns and calculate potential capital losses, utilizing **SQL**, Python, and Linux environments.
- Operationalized **Tableau dashboards** to visualize complex risk patterns, translating data findings for stakeholders to support banking and insurance risk mitigation.

SKILLS

- **Programming & Data:** Python (Pandas, NumPy, scikit-learn), SQL, Hive, ETL Pipelines, Data Modeling
- **Machine Learning:** End-to-End ML Pipeline Development, Supervised Learning, CatBoost, XGBoost, TF-IDF Vectorization, Model Evaluation
- **Analytics & Experimentation:** A/B Testing, Champion-Challenger Framework, Cohort Analysis, Causal Inference, North Star & Guardrail Metrics, Fraud & Risk Analytics, EDA
- **AI & Automation:** Claude, Model Context Protocol (MCP), Agentic AI Workflows
- **Visualization & Reporting:** Tableau, Power BI, Matplotlib, Seaborn, Microsoft Excel

EDUCATION

- Master of Science in Computer Science - **California State University, East Bay**, Hayward, CA Feb 2020 – Dec 2021
- Master of Technology in Computer Science - **Christ University**, Bengaluru, India Aug 2014 – Jan 2016

ACCOMPLISHMENTS

- **Employee Recognition Award:** Top-quality analytics deliverables at AIG Analytics and Services.
- **Top Performer Award:** Exceeded goals at TikTok, improving detection accuracy and user experience.